

|                                                         |                                                         |
|---------------------------------------------------------|---------------------------------------------------------|
| Program: <b>Diploma in Cloud Computing and Big Data</b> |                                                         |
| Course Code : <b>4278</b>                               | Course Title: <b>Data Structures and Algorithms Lab</b> |
| Semester : <b>4</b>                                     | Credits: <b>1.5</b>                                     |
| Course Category: <b>Program Core</b>                    |                                                         |
| Periods per week: <b>3 (L:0 T:0 P:3)</b>                | Periods per semester: <b>45</b>                         |

### Course Objectives:

- Implement Recursive Algorithms in Python
- Implement different Searching and Sorting Algorithms
- Implement linear data structures in Python like Array, Stack, Queue, and Linked List
- Implement Non-linear Data Structures in Python like Tree and Graph.

### Course Prerequisites:

| Topic                                                         | Course code | Course name                         | Semester |
|---------------------------------------------------------------|-------------|-------------------------------------|----------|
| Sound knowledge in Python Programming                         | 1008        | Introduction to IT systems Lab      | I        |
|                                                               |             | Programming in Python               | III      |
| Control Structures, Modular Program Design, Arrays, Functions | 2131        | Problem Solving and Programming     | II       |
|                                                               | 2139        | Problem Solving and Programming Lab | II       |

### Course Outcomes:

On completion of the course, the student will be able to:

| CO <sub>n</sub> | Description                                                           | Duration (Hours) | Cognitive Level |
|-----------------|-----------------------------------------------------------------------|------------------|-----------------|
| CO1             | Demonstrate recursive functions and array based programming in python | 8                | Applying        |
| CO2             | Implement Searching and Sorting Algorithms in Python                  | 10               | Applying        |

|     |                                                                                   |    |          |
|-----|-----------------------------------------------------------------------------------|----|----------|
| CO3 | Construct Stack, Queue, and Linked List data structures                           | 11 | Applying |
| CO4 | Illustrate Basic operations in non linear data structures such as Trees and Graph | 13 | Applying |
|     | Lab Exam                                                                          | 3  |          |

### CO – PO Mapping

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1             | 3   |     |     | 3   |     |     |     |
| CO2             | 3   |     |     | 3   |     |     |     |
| CO3             | 3   |     |     | 3   |     |     |     |
| CO4             | 3   | 3   | 3   | 3   |     | 3   |     |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

### Course Outline

| Module Outcomes | Description                                                                  | Duration (Hours) | Cognitive Level |
|-----------------|------------------------------------------------------------------------------|------------------|-----------------|
| CO1             | <b>Demonstrate recursive functions and array based programming in python</b> |                  |                 |
| M1.01           | Construct programs in python using recursive functions .                     | 3                | Applying        |
| M1.02           | Implement Array based sequences in Python                                    | 2                | Applying        |
| M1.03           | Illustrate Programs using Two-Dimensional Arrays in Python                   | 3                | Applying        |
| CO2             | <b>Implement Searching and Sorting Algorithms in Python</b>                  |                  |                 |
| M2.01           | Demonstrate Linear Search and Binary Search in Python                        | 3                | Applying        |

|            |                                                                                                                                                          |     |          |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| M2.02      | Experiment Bubble Sort, Selection Sort, Quick Sort and Merge Sort in Python                                                                              | 7   | Applying |
|            | Lab Exam – I                                                                                                                                             | 1.5 |          |
| <b>CO3</b> | <b>Construct Stack, Queue, and Linked List data structures</b>                                                                                           |     |          |
| M3.01      | Demonstrate Stack and Queue using Array in python                                                                                                        | 2   | Applying |
| M3.02      | Use stack for Infix to Postfix conversion and Postfix Evaluation in python                                                                               | 3   | Applying |
| M3.03      | Illustrate Singly Linked List operations - insertion, deletion, searching and traverse. in Python                                                        | 3   | Applying |
| M3.04      | Construct Stack and Queue using Linked List in python                                                                                                    | 3   | Applying |
| <b>CO4</b> | <b>Illustrate Basic operations in non linear data structures such as Trees and Graph</b>                                                                 |     |          |
| M4.01      | Demonstrate representation of Graph using linked list in python                                                                                          | 3   | Applying |
| M4.02      | Implement Graph Traversal Algorithms in Python                                                                                                           | 3   | Applying |
| M4.03      | Construct binary search tree and perform operations such as creation, insertion, deletion, traversal, count number of nodes and height of tree in python | 7   | Applying |
|            | Lab Exam – II                                                                                                                                            | 1.5 |          |

**Text /Reference:**

| <b>T/R</b> | <b>Book Title/Author</b>                                                                                                                          |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| T1         | Goodrich, Michael T., Roberto Tamassia, and Michael H. Goldwasser. <i>Data structures and algorithms in Python</i> . John Wiley & Sons Ltd, 2013. |
| R1         | Bradley N Miller Problem Solving with Algorithms and Data Structures Using Python, Franklin Beedle & Assoc; Second edition (1 January 2013)       |

**Online Resources:**

| Sl No | Website Link                                                                                                                                                                                          |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | <a href="https://www.tutorialspoint.com/python_data_structure/python_data_structure_introduction.htm">https://www.tutorialspoint.com/python_data_structure/python_data_structure_introduction.htm</a> |
| 2     | <a href="https://www.geeksforgeeks.org/python-data-structures/">https://www.geeksforgeeks.org/python-data-structures/</a>                                                                             |
| 3     | <a href="https://www.edureka.co/blog/data-structures-in-python/">https://www.edureka.co/blog/data-structures-in-python/</a>                                                                           |
| 4     | <a href="https://realpython.com/python-data-structures/">https://realpython.com/python-data-structures/</a>                                                                                           |
| 5     | <a href="https://www.programiz.com/dsa">https://www.programiz.com/dsa</a>                                                                                                                             |